# LPC1768 + HC-SR04 | COMPLETE LAB CONNECTION MANUAL

Exact equipment: ALS-SDA-ARMCTXM3-01 evaluation board (LPC1768), HC-SR04, breadboard, Dupont jumper wires, TWO 1 kOhm resistors and ONE 2.2 kOhm resistor. Optional on-board 16x2 LCD and buzzer.

**READ BEFORE CONNECTING: The manufacturer confirms the numbered electrical pin assignments, but this document cannot verify the physical pin-1 orientation, your actual jumper fit, or the condition of the lab board. DO NOT energize the sensor until the CN6 pin numbering, resistor values, and supply voltage have been checked.**

## A. What you need on the bench

| Item | Quantity / purpose |
|---|---|
| ALS LPC1768 board and its lab 5 V power supply | 1; supplied by lab |
| HC-SR04 ultrasonic module | 1; four pins VCC, TRIG, ECHO, GND |
| Breadboard | 1; standard center split, five-hole strips |
| 1 kOhm resistor | 2; R1 and R2 in series on upper part of divider |
| 2.2 kOhm resistor | 1; R3 from divider output to ground |
| Dupont jumpers | At least 6 usable connections; verify CN6 compatibility |
| 10-pin FRC ribbons | 2 for LCD and buzzer; normally lab board accessories |
| USB-serial / grey RS232 cable | Lab equipment for UART0 ISP |
| Multimeter | Borrow from lab for resistance and voltage checks |
| 26-pin IDC breakout / compatible adapter | Only if Dupont wires cannot reach CN6 contacts securely |

## B. Board connector map: use ONLY these positions

| Board contact | Signal | Purpose |
|---|---|---|
| CN6 pin 1 | P2.0 | HC-SR04 TRIG |
| CN6 pin 2 | P2.1 | Divider OUTPUT (Echo input), never direct 5 V Echo |
| CN6 pin 25 | +5 V via JP1 | Sensor VCC via breadboard + rail |
| CN6 pin 26 | GND | Sensor and divider common ground |
| CND -> CNAD | P0.23-P0.28 | LCD 4-bit data and controls via 10-pin ribbon |
| CNA -> CNA5 | P0.4-P0.11 | Buzzer interface; CNA5 pin 8 corresponds to P0.11 |
| CN9 left DB9 | UART0 RS232 | Flash Magic serial interface |
| CN8 right DB9 | 5 V power input | Use the lab-supplied power cable |

CN6 numbering is from ALS manual printed pages 25-26. Do not confuse CN6 (26-pin NIFC) with CND (10-pin GPIO). The CN6 pin-1 location is NOT established by the electrical table alone.

## C. Before touching any wiring: pin-1 and resistor verification

1. Switch board power OFF and unplug its power cable. Never rewire with power applied.

2. Locate the NIFC block and its CN6 26-pin boxed header on the board; distinguish it from CN5 JTAG, CN7 trace and CN8 power.

3. Find a molded triangle, square pad, printed 1 or cable key for CN6 pin 1. Confirm orientation with the lab instructor or the kit placement drawing; do NOT infer pin numbering from a generic 2x13 illustration.

4. Using that confirmed numbering, locate pins 1, 2, 25 and 26. Write them on masking tape beside the header (not on contacts).

5. Check that the actual wires or a breakout adapter make secure isolated connections to those contacts. If the connector is recessed and wires do not fit, STOP; obtain a proper IDC breakout.

6. Confirm R1 and R2 are each 1 kOhm and R3 is 2.2 kOhm using a meter (or verified color code). Never measure resistance in a powered circuit.

## D. Breadboard anatomy: essential to avoid accidental shorts

On a conventional breadboard, A-E of the SAME numbered row are connected; F-J of that row are connected separately across the center trench. Different numbered rows are isolated. Side power rails may be split partway along their length. In the steps below use the LEFT half (A-E) only for divider nodes, and use the + / - rail after checking continuity.

Example node layout (LEFT half): row 10 = Echo input; row 15 = junction A; row 20 = divider output B. A resistor must have its two legs in DIFFERENT numbered rows. Do not place both legs in A10 and E10: that would short the resistor.

## E. Exact resistor circuit using ALL THREE resistors

```
HC-SR04 ECHO (nominal 5 V pulse)
      |
   R1 = 1 kOhm
      |
   Node A  [row 15]
      |
   R2 = 1 kOhm
      |
   Node B  [row 20] ----> CN6 pin 2 / P2.1 INPUT
      |
   R3 = 2.2 kOhm
      |
  GND rail --------------> CN6 pin 26 / GND
```

**Divider math: 5.0 x 2.2 / (1.0 + 1.0 + 2.2) = 2.62 V at Node B (nominal Echo HIGH). This uses the exact three resistors you reported. Node A is about 3.81 V: DO NOT connect P2.1 to Node A.**

## F. Exact numbered breadboard installation

| Step | Put this component / wire here | Why |
|---|---|---|
| 1 | Choose left-half rows 10, 15 and 20 (A-E). Leave them otherwise empty. | Three separate nodes |
| 2 | Place R1 1 kOhm from A10 to A15. | Echo -> Node A |
| 3 | Place R2 1 kOhm from B15 to B20. | Node A -> Node B |
| 4 | Place R3 2.2 kOhm from C20 to the GND rail. | Node B -> ground |
| 5 | Connect HC-SR04 ECHO to B10 (same strip as A10). | Echo into R1 |
| 6 | Connect a jumper from D20 to CN6 pin 2. | 2.62 V Echo signal to P2.1 |
| 7 | Connect CN6 pin 26 to breadboard GND rail. | Common ground |
| 8 | Connect HC-SR04 GND to that SAME GND rail. | Sensor ground |
| 9 | Connect CN6 pin 25 to breadboard +5 V rail (JP1 short). | Sensor supply |
| 10 | Connect HC-SR04 VCC to that SAME +5 V rail. | Sensor VCC |
| 11 | Connect HC-SR04 TRIG directly to CN6 pin 1. | Trigger from P2.0 |

The exact letters A/B/C/D in this example are chosen only to identify distinct holes on the same five-hole row; any hole A-E of a specified row is electrically equivalent. If a sensor is inserted directly into the breadboard, place EACH of its four pins in a different numbered row and then route wires from those rows. Do not let two sensor pins share one row.

## G. Sensor pin order: read the PCB printing

| Viewing the sensor | Pin order (top to bottom, pins on left) |
|---|---|
| Front, round transducers facing you | VCC, TRIG, ECHO, GND |
| Back, ICs facing you | GND, ECHO, TRIG, VCC |

These directions match your two HC-SR04 photographs. Follow the labels printed next to the pins if the module is rotated.

## H. Add the on-board LCD - do not hand-wire its six data signals

| Step | Physical action | Signal / setting |
|---|---|---|
| 1 | Board OFF. Locate GPIO header CND and LCD header CNAD. | These are different 10-pin connectors. |
| 2 | Connect one straight-through 10-pin FRC cable CND -> CNAD. | Align pin 1 on both ends. |
| 3 | Short JP16. | Select 16x2 LCD (not graphic LCD). |
| 4 | Short JP5. | Connect P0.26 to LCD data line D7. |
| 5 | Leave graphic-LCD-only connections unused. | Do not connect CNA to CNAD for this 16x2 configuration. |
| 6 | After power-on, gently adjust POT3 with a screwdriver if characters are invisible. | Contrast only; do not force the potentiometer. |

| CND pin | LPC1768 GPIO | LCD function (through ribbon) |
|---|---|---|
| 1 | P0.23 | D4 |
| 2 | P0.24 | D5 |
| 3 | P0.25 | D6 |
| 4 | P0.26 | D7 (JP5 short) |
| 5 | P0.27 | RS |
| 6 | P0.28 | EN |

The six LCD mappings are derived from the manufacturer tables for CND and CNAD, assuming a straight 1:1 FRC cable. LCD supply and contrast are provided on-board; do not add a separate 5 V jumper to the LCD.

## I. Add the on-board buzzer

| Step | Physical action | Signal / caution |
|---|---|---|
| 1 | Board OFF. Locate GPIO header CNA and motor/relay/buzzer header CNA5. | Do not confuse CNA5 with CNA1 LEDs. |
| 2 | Connect straight-through 10-pin FRC cable CNA -> CNA5, pin 1 aligned. | Existing buzzer driver is on the board. |
| 3 | Firmware controls P0.11 as GPIO output. | CNA pin 8 -> CNA5 pin 8 -> buzzer. |
| 4 | Keep motor and relay loads disconnected. | Other CNA5 pins control those circuits. |

The buzzer input path is documented; the active-high/active-low behavior and acoustic pattern should be confirmed with the manufacturer buzzer test or a simple GPIO test before integrating the full application.

## J. Jumper and switch setup for UART0 ISP flashing

| Item | Required position | Purpose |
|---|---|---|
| JP1 | SHORT | +5 V to CN6 pin 25 |
| JP2 | OPEN | UART ISP, not JTAG |
| JP3 | SHORT | P2.10 ISP entry |
| JP5 | SHORT | 16x2 LCD D7 |
| JP6 | SHORT | Normal 3.3 V board supply |
| JP16 | SHORT | 16x2 LCD selection |
| SW21 | ON for flashing; follow lab procedure for running | UART0 DTR/RTS link; some example programs require OFF during UART use |
| Other jumpers | Leave in existing lab-approved configuration | Do not change unrelated CAN, USB, SPI, RTC jumpers |

## K. Physical PC + power cables (do this while board OFF)

1. Connect the lab USB-to-serial adapter to the computer USB port.

2. Connect its DB9 serial end to the long grey RS232 cable as shown in the MIT connection sheet.

3. Connect the long grey RS232 cable to the LEFT DB9 female UART0 connector CN9 on the kit.

4. Connect the lab-supplied grey power cable to the RIGHT DB9 male power connector CN8. Never use the RS232 cable as the power cable.

5. Check all ribbon connectors and breadboard wires again before applying power.

## L. Power-on checks: test without risking the GPIO input

1. With power OFF, leave the Node B -> CN6 pin 2 jumper DISCONNECTED for the initial electrical test. Also keep sensor VCC disconnected until the 5 V rail is confirmed.

2. Ask lab staff to verify TP1 ~5 V, TP2 ~3.3 V and TP3 GND, and that CN6 pin 25 to pin 26 measures ~5 V when JP1 is shorted. Power OFF again before connecting sensor VCC.

3. With power OFF, connect sensor VCC and complete all divider wiring, but keep Node B -> CN6 pin 2 disconnected. Power ON and measure Node B relative to ground while the Echo line is HIGH. Echo is pulsed, so a basic multimeter may not catch its peak; use an oscilloscope or logic analyzer if available.

4. Confirm Node B cannot exceed 3.3 V for the intended 5 V Echo input (nominal peak ~2.62 V). Power OFF; only then connect Node B -> CN6 pin 2.

5. Power ON and run a simple ultrasonic test before enabling LCD and buzzer outputs. If anything heats, smells, or behaves unexpectedly, switch OFF immediately.

**A multimeter reading of 0 V on Node B while the sensor is idle is normal and does NOT prove the divider works under a HIGH Echo pulse. Never claim that a static 0 V measurement verifies the peak.**

## M. Keil / Flash Magic: exact operational sequence

1. In Keil uVision select the NXP LPC1768 device and use the project startup/system files supplied for your kit. Include LPC17xx.h / CMSIS as required by the project.

2. Compile the program and resolve errors. Enable HEX file generation in project output settings; note the exact .hex path.

3. Keep the kit OFF while changing jumper and cable connections. For UART ISP use JP2 open, JP3 short, SW21 ON.

4. Power the kit ON. Open Flash Magic, select LPC1768, the actual Windows Device Manager COM port and the lab-supported baud rate. The MIT sheet recommends 115200; 9600 or 7200 may be needed for troubleshooting.

5. Erase flash as required by the lab procedure; select the new HEX and program it. Check the tool reports success.

6. To run the application, use the lab-approved reset / ISP exit procedure (typically remove JP3 while power OFF, then power ON / reset). Do not assume firmware runs while P2.10 is held in ISP mode.

7. If the program does not run, first check the correct HEX, reset/ISP state, 5 V and 3.3 V supplies, cable pin alignment and the actual GPIO assignments.

## N. Firmware pin assignments for the complete demo

| Module | Pin(s) | Firmware behavior |
|---|---|---|
| HC-SR04 TRIG | P2.0 / GPIO2 bit 0 | Output LOW, pulse HIGH for >=10 microseconds |
| HC-SR04 ECHO | P2.1 / GPIO2 bit 1 | Input; measure HIGH pulse width with Timer0 |
| LCD D4-D7 | P0.23-P0.26 | 4-bit LCD data bus |
| LCD RS, EN | P0.27, P0.28 | LCD command/data and enable strobe |
| Buzzer | P0.11 | GPIO output through on-board driver |
| UART0 programming | P0.2/P0.3 hardwired | Use Flash Magic via CN9 |

**IMPORTANT: The earlier ultrasonic-only C program did NOT implement LCD or buzzer output. Do not flash it expecting the display or buzzer to work. A complete application must initialize P0.23-P0.28, implement HD44780-compatible LCD timing, and drive P0.11 appropriately. Timer0 microsecond timing also depends on actual clock configuration.**

## O. Expected behavior and quick fault isolation

| Symptom | First checks |
|---|---|
| No power / no board LED | CN8 power cable, JP6, TP1/TP2/TP3, supply |
| Flash Magic cannot communicate | Correct COM port, CN9 serial, SW21, JP3, JP2, baud; consult MIT troubleshooting sheet |
| Ultrasonic always times out | P2.0 TRIG, P2.1 Echo via Node B, sensor 5 V/GND, divider continuity, correct PINSEL4 |
| Distance unstable | Rigid sensor mounting, target alignment, >60 ms trigger spacing, median filtering |
| LCD blank | CND -> CNAD, JP16 short, JP5 short, POT3, correct LCD firmware |
| Buzzer silent | CNA -> CNA5, P0.11 output, buzzer polarity/test program |
| Unexpected motor/relay activity | Stop: P0.4-P0.10 may be driven when CNA -> CNA5 is installed |

## P. Final before-power checklist

[ ] Board OFF while wiring
[ ] CN6 pin 1 confirmed physically; pin 2 / 25 / 26 identified
[ ] CN6 connection method mechanically secure
[ ] R1 = 1k, R2 = 1k, R3 = 2.2k confirmed
[ ] R1: row 10 -> row 15; R2: row 15 -> row 20; R3: row 20 -> GND
[ ] P2.1 jumper connects to row 20 (Node B), NOT row 15
[ ] +5 V and GND are not shorted
[ ] HC-SR04 VCC / TRIG / ECHO / GND verified from its printed labels
[ ] JP1, JP5, JP6, JP16 short; JP2 open and JP3 short for ISP
[ ] LCD CND -> CNAD and buzzer CNA -> CNA5 ribbons aligned
[ ] Serial to LEFT CN9; power to RIGHT CN8
[ ] Board supply verified by lab multimeter; GPIO input connected only after divider check

## Q. Sources and confidence

Primary: Advanced Electronic Systems ALS-SDA-ARMCTXM3-01 User Manual, hardware description and connector tables (printed pp. 19, 24-27, LCD and buzzer sections); MIT Manipal LPC1768 Connection Guide, Guidelines and Troubleshooting sheets; supplied ALS pin-and-wiring reference and photographs.
Documented: connector-number to MCU-signal assignments, JP1/JP5/JP6/JP16 functions, LCD and buzzer header pairings, board power/serial ports. Derived: 1:1 ribbon signal mapping and exact breadboard hole allocation (user-chosen layout). Not independently established: CN6 physical pin-1 orientation on your particular unit, resistor identity before measurement, actual supply voltage, adapter compatibility, and firmware function on your board.